

MICHAEL LEE

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EDUCATION

Colorado School of Mines — Golden, CO

Expected May 2028

- M.S. in Mechanical Engineering (Concentration: Thermal, Fluid & Energy Systems)

Northwestern University — Evanston, IL

Jun 2025

- B.S. in Mechanical Engineering (Concentration: Aerospace Engineering)
- Courses: Propulsion, Compressible Flow, Heat Transfer, CFD, Thermal Systems Design, Mechanical Design, Manufacturing

PROFESSIONAL EXPERIENCE

Entegris — Colorado Springs, CO

Jul 2026 – Present

Metrology Engineer Intern/Co-op

- Interpret technical drawings and measurement documentation to run coordinate measuring machine (CMM) inspections and identify out-of-tolerance dimensions on critical components used in semiconductor manufacturing
- Design, fabricate, and integrate custom test fixtures using SolidWorks CAD and 3D printing, to enhance CMM measurement repeatability and accuracy
- Develop, test, and refine CMM programs for machined and injection-molded parts, enabling accurate dimensional verification to support product quality and process control

PROJECTS

Gear Reducer Transmission System

Mar 2025 – Jun 2025

- Designed and integrated shaft #1 in a 3-shaft, 56 kW gearbox transmission, reducing input speed from 2400 rpm to 337 rpm, by calculating gear forces and combined stresses to guide material selection and bearing sizing
- Sized a 35 mm diameter 1050 steel shaft by performing fatigue and yield failure analysis with Goodman criteria, achieving >10 million cycle life with 1.27 safety factor; selected bearings and verified L10 bearing life at operating speeds
- Streamlined design process by implementing equations in MS Excel to compute fatigue life, safety factors, gear forces, and bearing loads, enabling rapid evaluation of possible configurations under consistent assumptions
- Built complete Siemens NX CAD shaft assembly model and detailed part drawings with standard fits and tolerances, incorporating keyways and auxiliary components (end caps, gaskets, spacers) to ensure proper manufacturing and assembly

Aerospace Testbed Design and Integration

Jan 2025 – Jun 2025

- Designed a modular wing testbed for a high school summer aerospace course, defining architecture and requirements for four subsystems — avionics, wing manufacturing, wing-fuselage attachment, and launcher — enabling 30+ students to fabricate and test custom wing designs
- Developed avionics subsystem by creating circuit diagrams and Siemens NX CAD assemblies for electronics placement; tested data acquisition by processing sensor data in Python gathered from repeated ~5-second flights, achieving lift and drag coefficients within 10% of predicted values
- Managed 5-month project timeline using Gantt and RAM charts to assign and track responsibilities across 7-person team, resulting in on-time delivery of deliverables, including final report, bill of materials, fabrication guide, and presentation
- Documented subsystem designs, testing procedures, and performance data in a 135-page report, ensuring clarity and consistency for stakeholders

CFD Analysis of Rayleigh-Taylor Instabilities

Feb 2025 – Mar 2025

- Simulated Rayleigh-Taylor instabilities using ANSYS Fluent Volume of Fluid model to analyze effect of Atwood number on perturbation wavelength and growth rate
- Applied Python for FFT-based post-processing analysis and validated trends against linear stability theory; reduced growth-rate error by 5.7% using Richardson Extrapolation with mesh dependency of 2.1%
- Documented methodology, verification steps, and detailed error analysis in technical report to substantiate results
- to ensure accurate inventory counts

SKILLS

Engineering: Siemens NX (CAD, FEA, Thermal), ANSYS (CFD, FEA, Thermal), GD&T, SolidWorks CAD, Python, MATLAB

Fabrication: Mechanical Assembly, Hand & Power Tools, Manual Machining (Mill, Lathe), Electronics Soldering & Wiring